offline
handwriting
→ text

Handwritten Text Recognition using CRNN-BiLSTM

Offline handwritten word recognition with CNN features, bidirectional context, and CTC loss

CNN features | BiLSTM context | CTC decoding

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Prepared project copy. Historical scores have not been reproduced.

Why this problem matters

MOTIVATION

Handwritten Text Recognition turns word images into editable, searchable text. It is useful when information is trapped in scans, forms, archives, and handwritten notes.

- Historical document digitisation
- Automatic form and receipt processing
- Searchable notes and handwritten records
- Human-in-the-loop transcription support

Why segmentation-free?

- Letter boundaries are hard to define in cursive writing.
- Connected strokes and variable spacing make manual cutting brittle.
- CRNN + CTC learns directly from word-level labels.

Key message: the model learns to read the whole word as a sequence, instead of first cutting it into individual characters.

Historical results not reproduced; see docs/RESULTS.md.

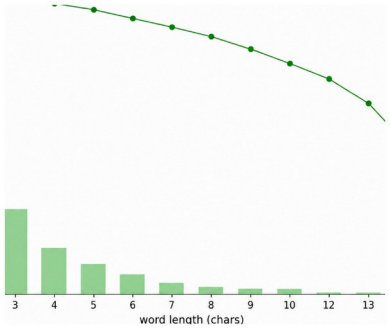
Recognition task

- Input: one grayscale word image resized to 32 × 256 pixels.
- Output: a variable-length character sequence representing the word.
- Training objective: learn from word-level transcriptions without manual character alignment.

Main challenge: longer or more cursive words create more opportunities for insertion, deletion, and substitution errors.

How performance is measured

Metric	Meaning
CER	Character error rate; lower is better
WER	Word error rate; lower is better
Exact accuracy	Full predicted word exactly matches label



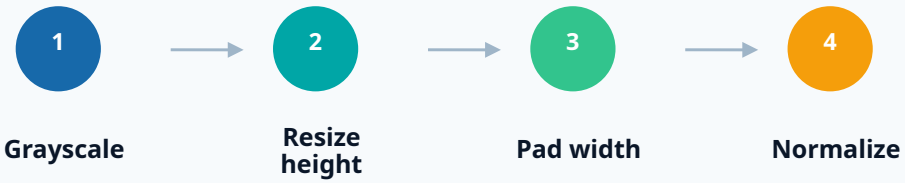
Historical results not reproduced; see docs/RESULTS.md.

Dataset and preprocessing

DATA SOURCE

- IAM word-level handwriting data, using word images and transcriptions.
- Labels are parsed from the words.txt annotation file.
- Invalid entries marked err and missing/corrupt images are removed.
- Project split: 80% train, 10% validation, 10% test.

PREPROCESSING PIPELINE



Final tensor format

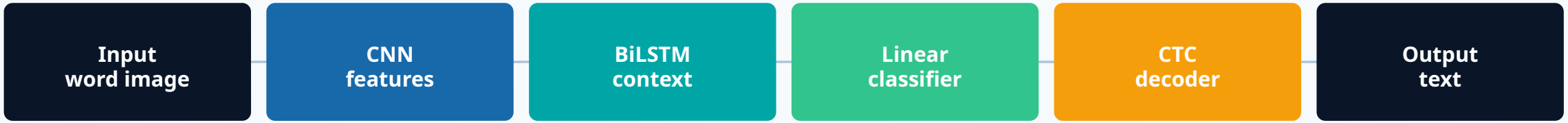
(B, 1, 32, 256)

Preserving aspect ratio reduces distortion. Padding keeps mini-batches efficient while labels remain variable for CTC.

Historical results not reproduced; see docs/RESULTS.md.

Model overview

END-TO-END SEQUENCE MODEL



1. Visual encoder

CNN scans the word image and converts local strokes into feature columns.

2. Sequence context

BiLSTM uses both past and future context to disambiguate characters.

3. Decoding

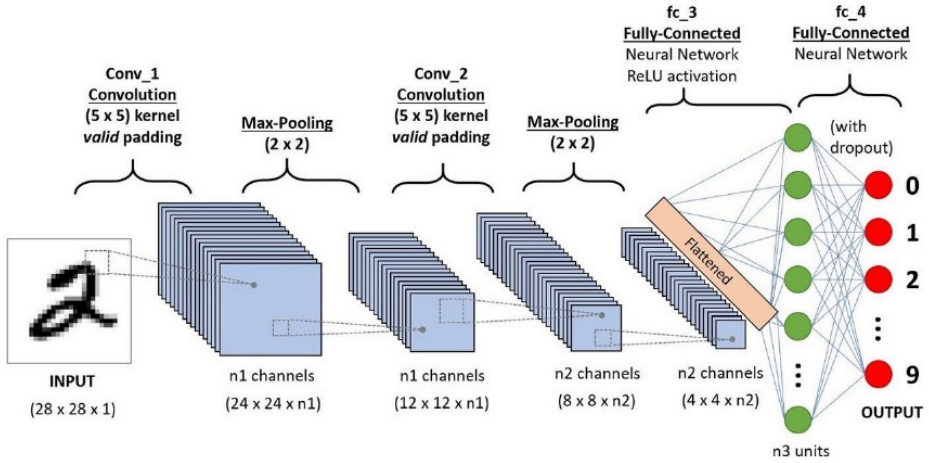
CTC collapses repeated symbols and blanks into the final word.

Historical results not reproduced; see docs/RESULTS.md.

CNN visual encoder

ARCHITECTURE DETAILS

- Seven convolutional blocks with ReLU activations.
- BatchNorm is used in the deeper 256- and 512-channel stages.
- Channel progression: 1 → 64 → 128 → 256 → 256 → 512 → 512 → 512.
- Pooling reduces height from 32 to 1 while preserving horizontal resolution.
- Final output is a sequence of 512-dimensional feature columns.



Interpretation: the CNN learns stroke-level and local shape features before the recurrent stage models the word context.

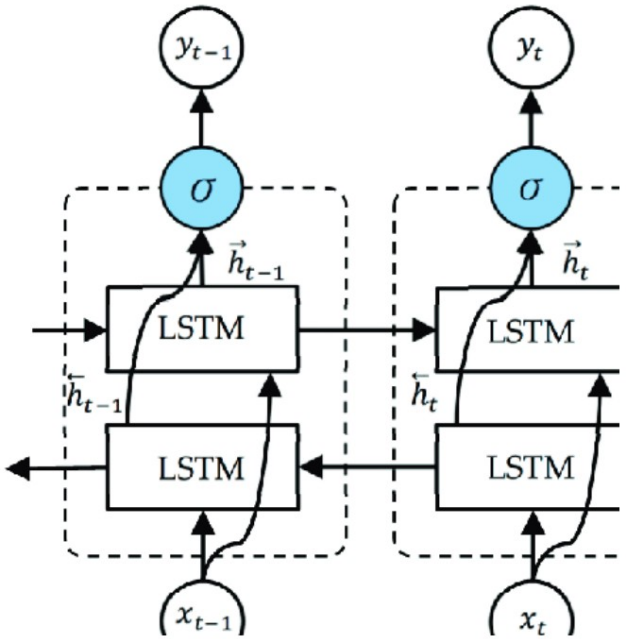
D = 512

T ≈ 63 for W=256

sequence features

Historical results not reproduced; see docs/RESULTS.md.

BiLSTM context and CTC decoding



BILSTM ROLE

- Two-layer BiLSTM with hidden size 256 per direction.
- Each position gets both left and right context.
- The concatenated state is 512-dimensional before classification.

CTC DECODING

- Learns from word-level labels without character-level alignment.
- Uses a blank token for no-character positions and repeated characters.
- Greedy decoding collapses repeats and removes blanks.

Example: time-step path “- h h e - a d - o n” → CTC collapse → “head-on”.

Historical results not reproduced; see docs/RESULTS.md.

Training setup

HYPERPARAMETERS REPORTED BY THE PROJECT

Component	Value
Input size	32 × 256 grayscale
Optimizer / learning rate	Adam / 1e-3
Batch size	32
Scheduler	ReduceLROnPlateau
Gradient clipping	5.0
Checkpoint criterion	Best validation CER

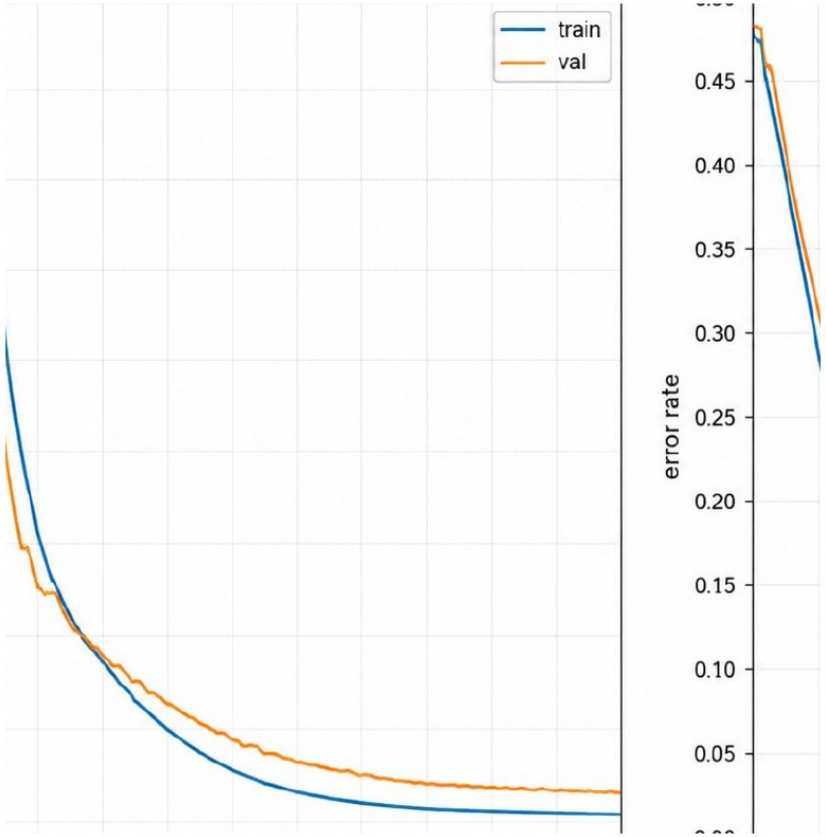
Training loop

- Forward pass produces logits with shape (T, B, C).
- CTC loss uses input lengths and target lengths.
- Backpropagation updates CNN, BiLSTM, and classifier together.
- CER and WER are computed after each epoch.

Accuracy note: report these as project-specific results; a standardized full IAM benchmark comparison would be needed for broader claims.

Historical results not reproduced; see docs/RESULTS.md.

Learning dynamics



WHAT THE CURVES SHOW

- Training loss decreases steadily across the 100-epoch run.
- Validation loss follows the same direction, with no clear divergence.
- CER and WER fall sharply in early epochs and then plateau.
- The final train/validation gap appears limited in the project curves.

Speaking point: convergence is stable, but the model still fails on ambiguous or long cursive words, so low loss does not mean perfect recognition.

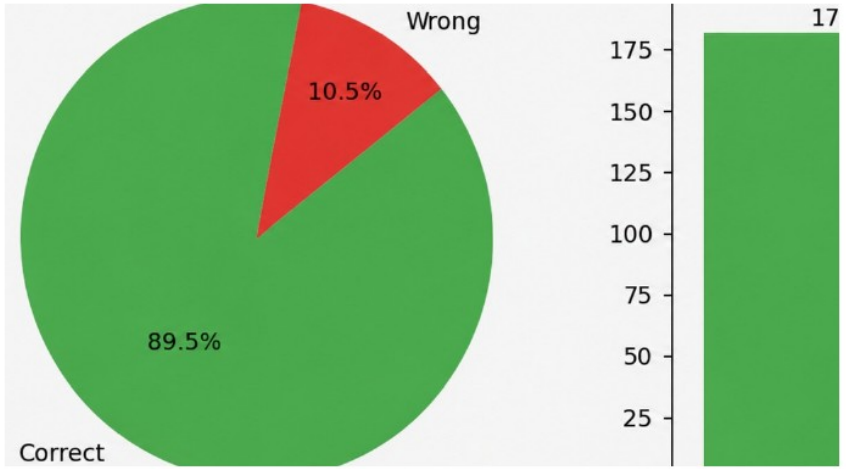
Historical results not reproduced; see docs/RESULTS.md.

Reported performance

0.012
Reported CER

0.027
Reported WER

89.5%
Exact accuracy



Interpretation

- Low CER means most individual characters are correct.
- WER is higher because one character error can make a whole word wrong.
- Exact accuracy is strict: 179 of 200 words in the reported held-out sample were correct.
- Use “reported project metrics” rather than claiming a universal benchmark score.

Bottom line: the model is reliable on clear word images, but exact recognition still depends on word length and handwriting clarity.

Historical results not reproduced; see docs/RESULTS.md.

Qualitative results and errors



WHAT SUCCEEDS

- Clearer words receive high confidence and are usually predicted correctly.
- Examples include words like “desolation”, “remained”, “message”, and “head-on”.

WHAT FAILS

- Cursive ligatures and strong slant make boundaries hard to infer.
- Some failures collapse into very short predictions when CTC is uncertain.
- Low confidence is a useful signal for manual review.

Practical takeaway: confidence can be used to decide which predictions need human checking.

Historical results not reproduced; see docs/RESULTS.md.

Limitations and next steps

CURRENT LIMITATIONS

- Evaluation is project-specific; broader benchmarking would strengthen the claim.
- Isolated word recognition does not yet solve full-line or paragraph transcription.
- Long words, cursive ligatures, and ambiguous glyphs remain the main error sources.

RECOMMENDED IMPROVEMENTS

- Beam search with a character- or word-level language model.
- Stronger augmentation: slant, affine distortion, elastic transforms, blur/noise.
- Full-line recognition and comparison with Transformer OCR models.

Final takeaway

CRNN-BiLSTM + CTC provides a compact, explainable baseline for offline word-level HTR.

Use it as a reproducible baseline: strong on clean isolated words, transparent about uncertainty, and extendable with beam search, language models, and line-level recognition.

Historical results not reproduced; see docs/RESULTS.md.